Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Software engineer with 8+ years shipping production systems and GTM automation - CRM automation, Salesforce integrations, and Clay/n8n workflow engineering at scale - plus hands-on agentic AI. Built **AgentProbe** (github.com/venkatalakshmiparimala/AgentProbe): a Langfuse-equivalent LLM observability and evaluation platform - 3-agent LangGraph runtime, 5-metric LLM-as-judge eval engine, CI-integrated regression gating. Built **Leasel**: 8-agent LangGraph HITL pipeline with 93.6% precision across 1,240 lease PDFs. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Designs, deploys, and measures AI systems that move real numbers.

CORE COMPETENCIES

GTM Automation & RevOps Engineering | LLM Evaluation Frameworks | Agent Orchestration (LangGraph, LangChain)

Salesforce · Clay · n8n Production Workflows | LLM APIs (OpenAI, Anthropic, Groq) | Python · TypeScript · JavaScript

Production AI Pipelines & SQL Optimization | FastAPI Backend Engineering

WORK EXPERIENCE

Entrata — PropTech / Real Estate SaaS

Sep 2018 – Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Built CRM data pipelines and GTM automation workflows** connecting Entrata's product data to commercial systems — Salesforce integrations (custom objects, reporting automations, data sync, API connections), Clay enrichment pipelines, and n8n orchestration; automated lead data flows and commercial operations at scale.
- **Shipped Entrata's first production AI pipeline** across 1.6M financial records (Python, PyTorch, OpenAI API) — LLM-powered automated data processing with output validation, quality gates, and compliance controls replacing manual Excel workflows.
- **Shipped 20+ full-stack production modules** (PHP, Python, PostgreSQL, JavaScript) across 120+ Agile sprints for 100K+ monthly active users — 99.5% uptime sustained over 6+ consecutive months; led sprint planning and story estimation.
- **Optimized 200+ SQL queries** across all client environments (30% performance gain) via execution plan tuning, indexing, partitioning, and caching; led database architecture decisions for complex production schemas.
- **Eliminated 80% of manual operational workload** by designing Python/Bash workflow automation pipelines; conducted 300+ peer code reviews and mentored 5+ junior engineers.
- **Achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines — zero security incidents; updated 50%+ of product codebase for encryption compliance.
- **Led microservices redesign** using Docker — 50% scalability improvement, 30% deployment time reduction; converted 20% of modules to service-based architecture.
- Managed end-to-end financial services (HAP allocation, rent pricing, HUD payments, tax credit calculations, income limits) across 100K+ unit portfolios in federal/state compliance.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern

- Built an agentic mission-planning system as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) delivering compliant vs. non-compliant routing with grounded rejection reasons.
- Implemented a RAG retrieval layer (FAA Part 107 + operational corpus) with cited-source answers — every compliance check functions as a hypothesis test grounded in source regulation.

George Mason University

Aug 2024 – Aug 2025

Research Associate / Graduate Teaching Assistant

- Achieved $R^2 = 0.977$ regression accuracy across 190+ countries by building multi-source prediction models (Python, R) on WHO and UK Road Safety datasets; applied Isolation Forest anomaly detection across 3 research pipelines.
- Designed faculty dashboards (Tableau, ggplot2, Quarto) improving research comprehension 40%; mentored 600+ students across Python programming and database design coursework.

Entrata India (Db Xento Systems)

Jan 2017 – Aug 2018

Associate Software Engineer / Trainee

- Built property management modules (income limits import across 1.6M HUD/Tax Credit records, document management, AWS S3 migration); migrated thousands of client documents resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: LLM Evaluation & Agentic AI Testing Platform

Live

Built a Langfuse-equivalent LLM observability and evaluation platform from scratch. Deployed a 15-endpoint FastAPI backend running 100% locally at zero cost with a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix), multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), and a CI-integrated 5-metric LLM-as-judge eval engine (factual accuracy, hallucination, retrieval, completeness, compliance) — every commit auto-scored, regressions blocked before merge. React dashboard with trace-level observability: per-run metrics, regression drift, and retrieval quality across concurrent agent test suites.

Python · LangGraph · LangChain · FastAPI · ChromaDB · React · Docker · GitHub Actions · OpenAI · Anthropic · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

Leasel: AI Document Intelligence & Support Investigation Platform

Built a dual-mode enterprise AI platform: (1) Document Intelligence — OCR-fallback ingestion, section-aware chunking, TF-IDF retrieval, confidence-scored answers with 100% citation correctness and human-review gate for regulated decisions; (2) Support Investigation — 8-agent LangGraph pipeline (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Human Escalation) with conditional escalation when confidence is too low. Achieved 93.6% precision and 87.9% F1 across 10 compliance rules on 1,240 residential lease PDFs.

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: NLP-Driven Medical Records Query System

Reduced manual clinical retrieval time ~60% and enforced HIPAA-compliant RBAC across structured medical records by building a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings.

Flask · React · LLMs · FHIR · Docker · spaCy · BERT · Embeddings — github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2017

SKILLS

GTM & RevOps: Salesforce, Clay, n8n, Zapier, CRM data pipelines, workflow automation

Languages: Python, TypeScript, JavaScript, SQL, PHP, R, Bash

AI & Agents: LangGraph, LangChain, LLM Orchestration, Tool Calling, ChromaDB, RAG Pipelines, LLM Evaluation, LLM-as-Judge, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, FastAPI

ML & Data: PyTorch, Scikit-learn, XGBoost, Isolation Forest, Random Forest, Feature Engineering, NumPy, Pandas, Tableau, Power BI

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Docker, GitHub Actions CI/CD, PostgreSQL, MySQL, Microservices, ETL, RabbitMQ